Name: _____ Date: _____ Period: _____

3-4: Where Are All the Roots?

TEKS P.5.J. Tags: **B** = Basic, **I** = Intermediate, **A** = Advanced.

Warm-up: Roots you already know

(a) Solve $x^2 - 5x + 6 = 0$ and write the factors: $(x - _____)(x - _____)$

(b) Solve $x^2 + 9 = 0$: $x =$ _____ or _____. Roots in all, for (a) and (b)? _____

Part A: Roots, factors, degree

Write every root, repeating a root as many times as its factor appears.

Factored polynomial	Degree	Every root (with repeats)	Roots in all
$(x - 3)(x + 4)(x - 6)$			
$x(x + 2)^2$			
$(x - 1)^2(x + 5)^3$			
$(x^2 + 9)(x - 2)$			

A polynomial of degree n has exactly _____ complex roots, counting repeats.

Part B: Do the i's cancel?

Multiply each pair. Use $i^2 = -1$.

(i) $(x - (2 + i))(x - (2 - i)) =$

(ii) $(x - (1 + \sqrt{2}))(x - (1 - \sqrt{2})) =$

What happened to the i's and the $\sqrt{2}$'s? If a + bi is a root of a polynomial with real coefficients, then _____ is also a root.

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Practice (real coefficients unless told otherwise)

1. B (a) Write a factor for each root: $-6 \to$ _____ $7 \to$ _____ $2/3 \to$ _____
(b) List the roots of $g(x) = (x + 1)(x - 8)(2x - 5)$: _____
(c) Dana says, "The root is 4, so a factor is $(x + 4)$." Substitute $x = 4$ to check her. What should she write?

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2. B (a) $f(x) = 3x^6 - x^2 + 1$ has exactly _____ complex roots, counting repeats.
(b) $g(x) = (x - 4)^2(x + 1)^3$ has degree _____. Every root, with repeats: _____
(c) Leo says, "g has only 2 roots, since there are only 2 different numbers." Agree? Explain.

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3. B Each list is roots of a polynomial with rational (so also real) coefficients. Write *every* root that must be there (forced partners and repeats), then the least degree.
(a) $5 + 2i$: _____ degree _____
(b) $-3i$ and 6: _____ degree _____
(c) $4 - \sqrt{3}$ and 2: _____ degree _____
(d) i, $1 + \sqrt{2}$, and 3: _____ degree _____
(e) 3 (a double root) and $2i$: _____ degree _____

#4–#6: Lowest-degree polynomial, real coefficients, leading coefficient 1. Factor first, then multiply to standard form.

4. I Roots 2 and $3 + i$.

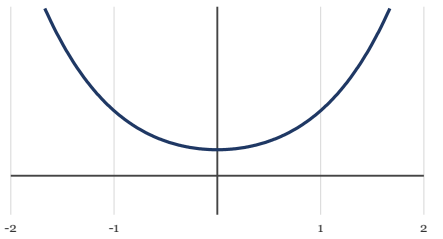
5. I Roots -1 and $2i$.

6. I Roots -3 and $1 + \sqrt{2}$ (rational coefficients).

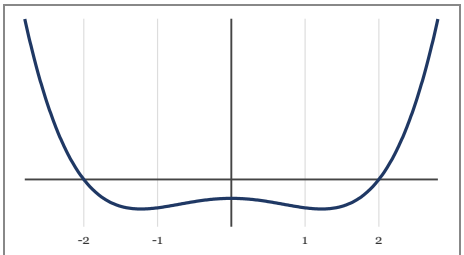
7. I Roots: 3 (a **double** root) and $1 - 2i$. State the degree, then write the polynomial in standard form.

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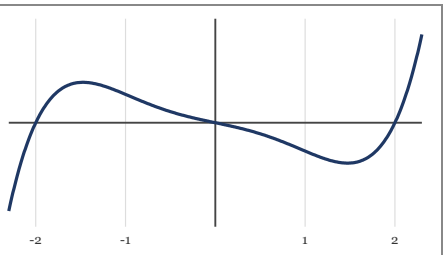
8. **A** Each graph is a polynomial with real coefficients, and every x-intercept is a simple crossing. Use the degree and the graph to complete the table, then match each graph to its equation.



Graph A (degree 4)



Graph B (degree 4)



Graph C (degree 5)

Graph	Degree	x-intercepts (x-values)	Real roots	Nonreal roots	Equation
A	4				
B	4				
C	5				

(i) $(x^2 - 4)(x^2 + 1)$ (ii) $x(x^2 - 4)(x^2 + 1)$ (iii) $(x^2 + 1)(x^2 + 4)$

9. **A** Real coefficients throughout.

(a) Could a degree 5 polynomial have a graph that never touches the x-axis? Use roots to explain.

(b) Write one degree 4 polynomial whose graph never touches the x-axis.

(c) Can a cubic have three nonreal roots? Can its roots be exactly 1, i, and 2i? Explain.

10. **A** A company's profit (thousands of dollars) from selling x hundred items is $P(x) = x^3 - 8x^2 + 22x - 20$, the polynomial from #4. Break-even means $P(x) = 0$.

(a) How many break-even points does the graph show? Find them.

(b) Where are the other roots, and what kind are they?

(c) Could another cubic profit model have no break-even point at all? Explain.

Math Time Machine: In 1629 Albert Girard wrote that an equation of degree n should have n roots if you count the "imaginary" ones. Gauss is usually credited with the first widely accepted proof, in his 1799 thesis. And in 1824 Abel showed that even though a degree 5 polynomial has 5 roots, there is no general formula built from roots to find them.